Surprising Findings

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SURPRISING FINDINGS

There are more things in heaven and earth, Horatio, than are dreamt of in your philosophy.

—WILLIAM SHAKESPEARE

AS I'VE TRIED TO CONVEY throughout this book, longitudinal studies are constructions of intrinsic contradiction and paradox. They require investment in massive information collection before there's any way to know for certain whether the information being compiled is the kind that can answer the questions the Study is posing. Much of this vast accumulation will almost assuredly never answer any questions at all. Yet in the huge heaps of data, little glints may sometimes be perceived. Some will turn out to be fool's gold. But some, suddenly—with a different cast of light or a sudden shift in context or a new analytical technology—identify themselves as twenty-four karat. There are unexpected, unexplainable, curious, and just plain odd findings in every large longitudinal study, and these are the very ones that beg for review every so often, just in case.

Here are some of the tantalizing glints from the Harvard Study of Adult Development. In time they may, as my final lesson from Chapter 2 suggests, help to elucidate one or two of life's enduring mysteries.

I. WHY DO THE RICH LIVE LONGER THAN THE POOR?

The last fifty years of epidemiology are making it ever clearer that the human lifespan is finite; after a point, the greatest riches in the world can't help you live longer. Nonetheless, the poor die sooner. In the United States it is socioeconomic status (income, education, occupation, access to health care) that is thought to account for this disparity, not the malnutrition and infection that shorten lifespans in third world countries.

Some people blame society for discriminatory access to health care and for toxic neighborhoods, poor nutrition, poor schools, and high unemployment; some blame the victim for dropping out of school, delinquency, bad habits, and poor self-care. While there is danger that the current emphasis on individual health promotion can be used in the service of victim-blaming, I've shown in Chapter 7 that the role of health-related behaviors cannot be carelessly dismissed out of political correctness. 1

Nevertheless, as Marcia Angell, former editor-in-chief of the New England Journal of Medicine, has pointed out, "Despite the importance of socioeconomic status to health, no one quite knows how it operates. It is, perhaps, the most mysterious of the determinants of health." 2

Nowhere is the Harvard Study of Adult Development as powerful as when it addresses this mysterious relationship between health and social class. The College and Inner City groups were matched for several important confounders: gender, race, geography, absence of delinquency, and a 1920–1930 birth cohort. But the two samples were clearly dichotomized by social class and intelligence (at least according to standard IQ tests), since the Inner City men were matched to a low-scoring group of delinquent youths. As Figure 10.1 illustrates, the Inner City cohort has been becoming disabled ten years earlier on average than the College sample, and dying ten years sooner; the estimated average longevity of the Inner City men is seventy years, of the College men, seventy-nine years.

The morbidity of the two samples is similar with regard to illnesses that are independent of self-care: that is, cancer (excluding lung), arthritis, heart disease, and brain disease. But there was twice as much lung cancer, emphysema, and cirrhosis, and three times as much Type II diabetes, among the Inner City men as in the College sample. The Inner City men were also more than three times as likely to be overweight. All these differences, however, disappeared for Inner City men who graduated from college.

Figure 10.1 Death after fifty for College men, Inner City men, and Inner City college graduates.

The average Inner City man was much less educated than the Harvard graduates, and he also led a far less healthy lifestyle. But the more education an Inner City man obtained, the more likely he was to stop smoking, avoid obesity, and be circumspect in his use of alcohol. The estimated average age of death for the Grant Study men who did not go to graduate school and the college-educated Inner City men was identical—seventy-nine years, if the World War II deaths are excluded.

So we have to ask: Does education really predict healthy aging independent of social class and intelligence? The college-educated Inner City men were neither more intelligent nor more privileged socially than their peers who did not attend college, so those two factors do not explain the nine-year difference in their estimated lifespans. The college-educated Inner City men had carefully tested IQs that were on average 30 points lower than their Harvard counterparts', and they attended lesser colleges. They were a full inch shorter, suggesting inferior childhood nutrition, and none of them had the upper-middle- or upper-class advantages typical of two-thirds of the Harvard men. In middle age, only half as many of the Inner City men as College men had made it into the ranks of the upper class, and they made only half as much money. Thus, neither intelligence nor status nor wealth can account for the disappearance of the nine-year shortfall in lifespan once an Inner City man had graduated from college. Parity of education alone was enough to produce parity in physical health.

Well-trained medical sociologists will scoff at this assertion. Do not the Whitehall Studies by Sir Michael Marmot in England appear to show that social class is one of the leading causes, if not the leading cause, of premature death in England? Did not the health of Whitehall civil servants improve with every step in their pay grade? 3 Yes. But Marmot's early Whitehall studies did not control for education or alcoholism. Our study of the Inner City men showed that education is very significantly associated with income and job promotion, and alcoholism is very significantly deleterious to both. 4 That is, with every step in pay grade, the chances rose of a man being nonalcoholic and better educated. It is likely that it is these factors, not the job or the pay grade, that facilitate better health. This is yet another instance where the choices we make appear to influence how long we will live.

Most social science studies, including Marmot's own early ones, control only for self-reported alcohol consumption, which, as I have been at pains to point out, correlates very poorly with objective alcohol abuse. In our samples, however, the difference in health outcomes among occupational levels diminished sharply once we controlled for alcohol abuse, which tends to depress occupational level. Alcoholism is bad for career advancement as well as for health.

The question then becomes: If education so powerfully affects self-care (as well as the more obvious job status), what affects whether or not people stay in school? In general, pursuit of education is most successful in coherent communities that invest heavily in their families and their school systems in an atmosphere of gender and racial tolerance. People with no hope for the future don't pursue education effectively. Providing that hope is the responsibility of the community, not the individual.

That said, however, the pursuit of education also reflects individual personality traits of perseverance and planfulness —traits that Friedman and Martin have shown to be important to longevity. 5 An important postscript to these findings is a recent paper by David Baber and colleagues, who find that it is not only increased education per se that reduces mortality; of equal importance was what Baber et al. called health reading fluency —the capacity to read prescription bottles, understand preventive services, and so on. 6

II. IS PTSD DUE TO COMBAT OR TO PERSONALITY DISORDER?

The frequency of posttraumatic stress disorder (PTSD) among returning Vietnam veterans has led many to wonder whether the principal causal factor is not severe combat stress after all, but pre-existing personality disorder. To address this question, the Grant Study took advantage of the fact that most of its members were World War II veterans. Not only had they all been extensively studied before the war, but they were all extensively debriefed after the war on their combat experiences, their physical symptoms during combat, and their persisting stress-related symptoms. John Monks, an internist particularly interested in combat experience, carried out these debriefings. 7 Forty years later, sociologist Glen Elder and I asked all the surviving College veterans (excluding the early Study dropouts) to fill out questionnaires regarding persisting symptoms of posttraumatic stress. (PTSD had not yet been "invented" in 1946, but its principal symptoms had been anticipated by the prescient Monks.) One hundred and seven men returned questionnaires. These men had also completed the NEO, which, as I've described in Chapter 4 , is an extensive and popular multiple-choice scale that includes a scale for the trait Neuroticism. 8 We were particularly interested to learn if the men who developed symptoms of posttraumatic stress had already looked vulnerable before the war, or if combat really was the primary factor in symptom development. (Please note here, however, that since the Study men had been protected to some degree by education and rank from the circumstances in which frank PTSD develops, we were studying symptoms of posttraumatic stress, not the disorder per se.)

First, the men who experienced the most intense combat did not appear to have been more vulnerable by nature; in fact, they manifested superior psychosocial health in adolescence and also at age sixty-five. Second, it was only the men with high combat exposure who continued to report symptoms compatible with PTSD after forty years. Third, we found that symptoms of posttraumatic stress both in 1946 and in 1988 were predicted independently by two factors: combat exposure and number of physiological symptoms during combat stress (but not during civilian stress). As I've already noted, severe combat exposure also predicted early death. It was noteworthy that the symptoms of posttraumatic stress reported in 1946 were not correlated with evidence of subsequent major depressive disorder, alcohol abuse, or poor psychosocial adjustment. Only combat exposure made a significant statistical contribution to posttraumatic stress symptoms, and that contribution was very significant. And while the NEO trait Neuroticism was associated with bleak childhood, psychiatrist utilization, poor psychosocial outcome at age forty-seven, and physiological symptoms during civilian stress, it was not associated with PTSD. 9

Sixteen men who endured severe combat reported no posttraumatic stress symptoms in 1946, and in 1988 still could not recall ever having had such symptoms. When we compared these sixteen resilient men to the eighteen with high combat exposure who did experience symptoms, their Neuroticism scores were the same. However, their defensive styles were not. The high combat veterans who manifested less mature defenses as young adults had very significantly more symptoms than those with more mature defenses (Chapter 8). Equally important, seven (39 percent) of the men with high combat experience and less mature defenses were dead by age sixty-five, while no man with mature defenses was. In the Grant Study, high combat exposure per se was not associated with a higher incidence of postwar alcoholism, but as I've said, the Grant Study men also did not develop full-blown PTSD.

The take-home messages here are: first, that combat exposure predicted symptoms of posttraumatic stress, but pre-existing psychopathology did not; and second, premorbid emotional vulnerability predicted subsequent psychopathology, but not symptoms of post-traumatic stress. Without the kind of prospective data and prolonged follow-up that the Grant Study made available, this finding would not have been possible.

III. POLITICS, MENTAL HEALTH, AND . . . SEX

As an optimist, a psychoanalyst, and a Democrat, I brought many prejudices to the Grant Study. I've always been delighted to have them confirmed, and this actually has happened occasionally. There is such a thing as a happy marriage, for example. And although colleagues derided for years my conviction that alcoholics could not return safely to controlled drinking, patience (as I detailed in Chapter 9) has seen me vindicated. 10

Still, the greatest value of longitudinal study is its way of shattering prejudices and superstitions, and the Grant Study has done a real job on some of mine. I held a deep belief, for example, that Republicans are neither as loving nor as altruistic as Democrats. But after it occurred to me that Gandhi was a very bad father and John D. Rockefeller rather a good one, I thought that I had better test that notion out.

My first step was to identify political preferences from 1950 to 1999. Every biennial questionnaire had asked about politics, and we knew a lot about the men's political views. We knew, for instance, that in 1954 only 16 percent had sanctioned the McCarthy hearings. We knew that like many liberal members of their generation, the Grant men were "for" equal rights, at least after the fact. They applauded the Supreme Court decisions and civil rights legislation after they happened, but only very few took an active role in trying to bring about racial or gender equality. In 1967, 91 percent were for de-escalating our involvement in Vietnam; this was true of only 80 percent of their classmates. Had the Grant Study subjects had their way, Eugene McCarthy and Nelson Rockefeller would have been nominated in 1968 rather than Humphrey and Nixon; Gore would have easily beaten Bush in 2000, and the U.S. would never have invaded Iraq without further consultation or U.N. permission.

With all this information, it was not hard for independent raters to score the men along a political continuum from 1 (very liberal) to 20 (very conservative). Let me illustrate these abstractions briefly. Bert Hoover received a 20 on our scale (the highest score possible) for political conservatism. In 1964 he was a Goldwater supporter who thought that America should plan an invasion of Cuba. In 1967, he advocated "dropping Lyndon and Hubert on Hanoi" to resolve the problem of Vietnam. He saw student protests (and hippies and drug users as well) as "evidence of a sick society, a culture that is overripe and ready to be clobbered by a more vigorous one." In 1972 Hoover disapproved of long hair, premarital sex, anti-war demonstrations, and marijuana, and said that he would forbid his children to date a black person. Asked in 1981 what blessings he perceived in a Reagan presidency, he almost shouted, "The country may survive. Especially if we can rid the House of idiot liberals!" The public figure in the last two hundred years that he most admired was Herbert Hoover; and in 1985 when asked to check his current political stance—conservative or liberal—he wrote in, "just left of Genghis Khan." In 1988, with regard to "student protests over disarmament and university divestment in South Africa," he circled "Disapprove" and inked in, for emphasis, "The dumb bastards." In 1995, he was asked to check "What do you think of Newt Gingrich?" with choices ranging from "Delight" to "An abomination." Hoover circled "Delight" and again ad-libbed for emphasis, "Sorry. The abomination [that is, Bill Clinton] is in the White House!" In 1999, Hoover nominated Ronald Reagan for Time 's Man of the Century.

Oscar Weil, on the other hand, got a score of 3 on the political continuum. He had returned far fewer questionnaires than Hoover had, but of the Study black sheep Weil was one of my favorites. In 1964 he had supported Johnson over Goldwater. In 1967, he suggested that the way to end the crisis in Vietnam was to "Halt all offensive operations; start drawing up a time-table with South Vietnam for return of U.S. forces [to U.S.]."

Weil did not buy the whole Sixties package. In 1972, asked how he felt about his children and long hair, he responded insightfully, "I know it is irrelevant, but somehow I can't cheer it on." He added, "I know it's silly, but I have difficulty with premarital sex." But he had no problem with his daughters dating black people, and about the anti-war demonstrations he wrote, "On the general level of protest, I am most excited. I think that this is a really revolutionary time in this country and holds great promise for a real leap forward." He supported Walter Mondale for president in 1984, and Michael Dukakis in 1988. With regard to student protests for disarmament and against university investment in South Africa, he checked "approve" and inked in for emphasis, "I am more sympathetic to purely symbolic acts than I once was." In 1992 he voted for Clinton "with pleasure." He voted for Clinton again in 1996.

Most of the men in the Study clustered toward these two poles of the political continuum, and most maintained their positions for fifty years. The distribution of scores along the liberal-conservative axis was not a bell curve; it looked more like the twin humps of a Bactrian camel. And when I ran my test of how politics correlated with love, nobility, and so forth, it proved that my lifelong convictions about the virtues of liberals had no validity at all. Democrats had no superiority over Republicans with regard to the stability of their marriages, their mental health, their altruism, or their aging. Conservatives and liberals enjoyed identical scores on the Decathlon of Flourishing, and there was no difference in their relationships with their children.

That didn't mean that there were no real (and significant) differences between liberal and conservative. Liberals were more likely to be open to new ideas, and to approve of the younger generation's behavior. They were more likely to have had highly educated mothers, to have gone to graduate school, to display creativity, and to use sublimation as a defense. Conservatives were less open to novelty, but they made more money, played more sports, and were twice as likely to be religious.

I wondered what influences might inform political alignment, and went back to the men's college records. The two political poles could easily be identified in college, but—at least in this cohort—choice of one or the other had nothing to do with the quality of the men's childhoods, and little to do with parental social class. Remember, Identity in the Eriksonian sense means disentangling your values, your allegiances, and your politics from your parents'.

Surprisingly, however, the men's personalities in college had everything to do with their politics at eighty. As I've recounted, in 1942 each man was rated on twenty-six personality traits. 11 Two of them, Pragmatic and Practical organizing —both consistent with hardheaded common sense—were not significantly correlated with many facets of mental health. But they were very significantly associated with conservatism half a century later. Conversely, five traits quite un correlated with mental health and not well regarded by the early Study staff (but epitomizing many people's stereotype of Harvard students)— Introspective, Creative and intuitive, Cultural, Ideational, and Sensitive affect —were very significantly predictive fifty to seventy years later of being classified as a liberal.

Before he died at eighty-six, my very Republican, very intolerant, and as best I can tell, very mentally healthy grandfather used to chaff me with his take on the old epigram: If you're not a socialist before you're thirty, you have no heart; and if you're still a socialist afterward, you have no brains. But as a good developmentalist and champion of prospective study, I can now see that he was misled by retrospection. He'd been a McKinley Republican even as a young man. The years of Grant Study questionnaires confirmed that political convictions tend to endure, and that most young liberals and conservatives evolve smoothly into older ones. William James wasn't all wrong.

At the end of the day, those differences were not important. Even if "openness" were to be counted as a virtue—as my liberal prejudices would have it—that trait did not correlate with successful aging. For a while I was prepared to say that political conservatism predicts nothing about the future but one's politics. But, as it turned out, there was one big surprise hidden in the men's political profiles. As I noted in Chapter 6 , the two early traits that predicted conservatism also predicted early sexual inactivity, and the five traits associated with being a liberal predicted sustained sexual activity fifty years later. The seven most liberal men (with scores of 0–5) did not on average cease sexual activity until after the age of eighty. The seven most conservative men ceased sexual activity at an average age of sixty-eight. That was a significant difference. I have consulted urologists about this; they have no idea why it might be so, but it is an interesting finding nevertheless. It was curious too that the variables I thought would protect against early impotence—ancestral longevity, maturity of defenses, physical health at age eighty, and quality of marriage—showed no effect, and the men who at eighty-five were still sexually active were eight times as likely to have been rated Shy, Sensitive, Ideational, and Self-conscious and Introspective as those who became sexually inactive before seventy-five.

IV. HEALTH AND RELIGION

The modern world seems to be of two minds with regard to religion. On one hand, in the last fifty years Sunday school attendance in the U.K. has dropped from 74 percent to 4 percent, and Oxford University evolutionary biologist Richard Dawkins can declare, "I think a case can be made that faith is one of the world's great evils, comparable to the smallpox virus but harder to eradicate." 12 On the other, Gallup polls point out that 85 percent of Americans believe in God, and an increasing number of researchers report a positive association between religious commitment and health. 13 Nevertheless, despite growing evidence that there is a relationship between health, religious beliefs, and participation in religious activities, investigators remain uncertain about what might account for it. Some critics suggest that the relationship between religious commitment and physical health may be illusory, due to inadequate control of confounding factors that undermine both physical health and involvement in religion. 14 Such confounders—such as, and especially, poorly matched control groups and inadequate assessment of alcohol and smoking history—have marred prior studies. A fifty-year-old church-going Mormon is almost certainly a lifelong abstainer from alcohol. That fact must be separated out from his religious beliefs and engagements as a factor in his physical health outcome. Here again, prospective study is essential.

The College cohort was ideally suited to explore such concerns. Participants were a well-studied, culturally and socioeconomically homogeneous sample of men who varied widely in religious involvement and in health habits, but who all had excellent access to education and health care. Starting in 1967, in the College men's twenty-fifth reunion questionnaires, I began investigating the depth of their reported religious faith and the extent of their participation in it. I wanted to test a hypothesis that religious interest would increase with age. Also, I suspected that religious involvement would make a positive contribution to aging, both from a medical and a psychosocial perspective.

I assessed religious involvement over a thirty-year period on a 5-point scale: 1 = no involvement; 3 = some involvement; 5 = deep involvement. By design, different questionnaires explored different aspects, including how frequently a man attended religious services, how important he considered a personal belief in God, how applicable he considered religious considerations to his own life, and so on. Ninety men (two-fifths) fell into category 1 (close to atheism), and forty-eight men (one-fifth) fell into category 5 (devoutly religious).

We also assessed interest in "private spiritual practices" and other markers of interest in individual spiritual development as separate from faith-based beliefs and institutions. In our sample, the two admittedly different concepts of religion and spirituality overlapped so markedly that it was not possible to find any significant outcome difference between their effects. Fewer than 3 percent of the men fell in the top third in the religion category and the bottom third in the spirituality category, or vice versa. In short, our measure of "religious involvement" appears to have good face validity in measuring interest in "spirituality" too.

Counter to my first hypothesis that religious involvement would increase with age, 58 percent of the men reported little religious involvement at sixty-five in contrast to only 28 percent in college. Nine (26 percent) of thirty-five men who had gone to church regularly when young had virtually no involvement after age sixty. However, fifteen (25 percent) of the sixty men with little religious involvement when young became very involved after age sixty. Why? One possible reason might be that those fifteen men manifested nine times as many symptoms of depression, and spent three times as many years disabled or dead before age eighty, as the nine men whose religious involvement declined over time. This doesn't mean that religion is bad for your mental state. But the ill are more likely to go to the hospital than the healthy, and the depressed and anxious are more likely than the psychologically robust to seek religious comfort. 15

Table 10.1 contrasts the associations and consequences of religious involvement with the associations and consequences of mental health. Associations included parental social class, warmth of childhood, psychosocial soundness in college, smoking (in pack years), and alcohol abuse. 16 I expected to find a positive relationship between physical health and religious involvement, but Table 10.1 reveals none. There were 224 men for whom we had adequate data about their participation in religion over the course of their lives. By 2012, 63 (70 percent) of the 90 men with no religious involvement had died, and 36 (75 percent) of the 48 men with deep involvement had died. Notably, in these two well-matched groups of agnostic and religiously committed men, not only was there no difference in mortality, there was also no difference in cigarette and alcohol abuse. Mormons, Seventh-Day Adventists, and evangelical Christians live longer than their agnostic neighbors; they abuse cigarettes and alcohol far less. This was not true, however, for the devout Grant Study Irish Catholics and Episcopalians. The likelihood is that abstinence from drinking and smoking play more of a role in increased longevity than religious devotion itself.

Table 10.1 Religious Involvement and Adult Adjustment

Very significant = p<.001; Significant = p<. 01

Depression and an increase in stressful life events correlated significantly with increased religious involvement and also with psychiatric visits, but psychiatric visits and religious involvement were not correlated with each other except that they both appear to reflect independent ways that the men coped with life's difficulties. And if religion was not important to the longevity of the men in general, it was useful to the forty-nine men with the highest "misery quotient," including eighteen men who suffered from major depressive disorder, twenty-five who had experienced multiple stressful life events not of their own making, and six men of whom both these conditions were true. 17 Among the ten "miserable" agnostics, only one survived until age eighty-five. But among the fifteen "miserable" but deeply religious men nine were still alive—an almost significant difference (p<.02). Looked at from a psychosocial perspective rather than a strictly medical one, religious devotion remains one of humanity's great sources of comfort, and for many denominations replaces the use (and abuse) of alcohol and cigarettes.

My last hypothesis—that religious involvement would be associated with enhanced social support (as a consequence of more loving behavior toward others)—was not confirmed. The items in Table 10.2 were selected from two personality questionnaires, one administered when the men were about fifty years of age, and one when they were seventy-five. 18 The first four items were chosen as common reflections of religious faith, and the second three as reflections of real-life loving relationships. All items used a True = 1, False = 0 format.

Table 10.2 Spirituality and Warm Relationships

Very significant = p<.001; Significant = p<. 01; NS = Not significant.

The items chosen to reflect spirituality did correlate highly with religious involvement, but not with adult adjustment from age fifty to sixty-five. Conversely, the second group of items correlated significantly with adult adjustment, but not with religious involvement. 19 To embrace the last three items requires a happy childhood or a loving spouse. The first four items do not; they can be honored in the abstract, whatever one's life circumstances or relational capacities. More, for people whose circumstances or capacities do not facilitate loving real-life relationships, they promise other sources of contact. Even if a bleak childhood has taught you not to trust, or a major depression has distanced your friends, God continues to love you. Religion is a source of comfort for people whose concrete sources of love are limited. Beethoven saved himself in an angry depression by writing incomparable faith-inspired music, and he set to it Schiller's words: "Be embraced, all ye millions, with a kiss for all the world. Brothers, beyond the stars surely dwells a loving father."

There are some disparities between these Grant Study findings and those of other well-designed studies, and they deserve comment. An impressive body of research suggests that attendance at religious services protects against premature mortality. 20 But none of them show a direct causal effect, and too many rely on self-reports of alcohol abuse and physical health rather than on objective evidence. 21 As I've said, some deny the effects of alcohol and cigarettes entirely. One elegantly designed and analyzed study by leading social scientists notes that "church attendees were also less likely to smoke and drink, but those variables did not significantly affect mortality; thus they are not considered further." 22 Yet in virtually all studies conducted by physicians, alcohol abuse and smoking are among the most important causes of premature mortality. In the Grant and Glueck Studies, early mortality was four times higher among the alcohol and cigarette abusers than among nonsmoking social drinkers. 23 In our underprivileged Inner City sample, religious involvement at age forty-seven appeared to predict significantly fewer years of disability by age seventy. But the significance of this effect disappeared when smoking, alcohol abuse, and years of education were controlled for.

Another compelling explanation for our aberrant findings is that many American studies linking religious observance to physical health come out of the so-called Bible Belt, where agnostics are, at least statistically, also social outliers. In our sample of highly educated men centered in the northeastern United States, high religious involvement was not the cultural norm; more important, it was not deeply tied in to other sources of social supports. 24 In other words, in samples where healthy social adjustment usually includes clear religious involvement, such involvement is likely to correlate with warm relationships, social supports, and good physical health. Evidence of that correlation, however, does not reflect a direct causal relationship between religious involvement and health.

V. THE IMPORTANCE OF MATERNAL GRANDFATHERS

The mean age at death of the College men's grandparents (an 1860 birth cohort) was seventy-one years, and the mean age at death of their parents (an 1890 birth cohort) was seventy-six years. By historical standards, such ancestral longevity is remarkable, and comparable with that predicted for some contemporary European birth cohorts. 25 The likelihood of adventitious ancestral death from poor medical care, hazardous occupation, poor nutrition, or infection was less in the Grant Study sample than for less socioeconomically favored groups. This reduction in environmentally mediated mortality increased our chances to identify some genetically mediated effects on longevity.

While examining the effects of ancestral longevity on sustained good physical and mental health, we noted a marked and unexpected association between age at death of maternal grandfathers and the mental health of their grandsons. 26 The age of death of the other five first-degree ancestors (mother, father, maternal grandmother, and both paternal grandparents) was associated with virtually nothing in the men's lives except, modestly, with longevity. But the associations with the maternal grandfather's (MGF's) age of death were extraordinary. For example, while the average longevity of the other five ancestors was not relevant to the men's scores in the Decathlon of Flourishing, the maternal grandfathers of the men with the highest flourishing scores lived nine years longer than those of the men with the lowest scores—a significant difference. The average age of MGF death for 147 Study men who never saw a psychiatrist was seventy years. The average age of MGF death for the thirty-two men who made one hundred or more visits to a psychiatrist was sixty-one—a very significant difference—but the age of death of the other five relatives made no difference as to psychiatric visits. The maternal grandfathers of upper-class men lived only three years longer than those of men from blue-collar families.

In 1990, two board-eligible psychiatrists blind to other ratings were given the complete records of the sixty-one men in the Study who, by age fifty, manifested objective evidence of sustained psychosocial impairment (that is, psychiatric hospitalization, scoring in the bottom quartile of psychosocial adjustment at age forty-seven, or having used tranquilizers or antidepressants for more than one month). Many of these men were dependent on alcohol. They were rated for eight correlates of depression, the DSM-III criteria for major depressive disorder not yet having been developed. These eight correlates were (1) serious depression for two weeks or more by self-report, (2) diagnosis of clinical depression at some point in the man's life by a non-Study clinician, (3) use of antidepressant medication, (4) psychiatric hospitalization for reasons other than alcohol abuse, (5) sustained anergia or anhedonia, (6) neurovegetative signs of depression (e.g., early morning awakening or weight loss when depressed), (7) suicidal preoccupations, attempts, or completions, and (8) evidence of mania.

Of the sixty-one, thirty-six men were categorized as alcoholic or personality disordered. The remaining twenty-five included twelve men categorized by only one rater as having major depressive disorder and thirteen who both raters agreed manifested major depressive disorder. These twenty-five men met at least three—and an average of five—criteria of major depressive disorder. The other thirty-six men met an average of only one-tenth as many criteria.

As a contrast, we identified fifty undistressed men who through the age of sixty had never reported evidence of alcohol abuse, had made no visits to a psychiatrist, and had not used psychotropic drugs more than one day a year on average over twenty years. (This stipulation covered the contingency, say, of a man's having had to take Librium briefly during a surgical admission, or Ambien to recover from jet lag before an important overseas conference.) In addition, these men were classified in college as having well-integrated personalities, and the Study had never assigned them a psychiatric diagnosis. They were the antithesis of the depressed men.

Table 10.3 shows our four diagnostic groups: undistressed, alco-holic/personality disordered, major depressive disorder, and "intermediate"—that is, the men who didn't fall into any of the first three. The mean age at death of MGF for the twenty clearly depressed men was sixty. The mean age at death of the fifty-eight undistressed men was seventy-five—a very significant difference. The age of death of the MGF of the ten men with the highest anxiety scores on the NEO was fifty-seven, and that of the MGF of the ten men with the lowest anxiety scores was eighty-three, an even greater distinction.

This unexpected discovery—that of the six ancestors, it was only the maternal grandfather's longevity that was associated with affective disorder in the grandsons—is consistent with a linkage mediated via the X-chromosome. X-linked disorders such as hemophilia, color blindness, and baldness come through the mother's father, when it is he who supplies the grandson's only X chromosome. Such illnesses tend to skip the mother, who has a second, usually unaffected X chromosome to protect her. For half a century, researchers have speculated that there might be an X chromosome linkage in the etiology of de pression. 27 But this hypothesis has been confirmed only inconsistently in the search for a specific gene for bipolar disorder. 28 It seems clear that bipolar disorder and major depressive disorder are genetically heterogeneous—that the genes for the two disorders are different.

Table 10.3 Mean Age at Death of Maternal Grandfather and of Other Primary Ancestors in the Four Affective Disorder Categories

Very significant = p<.001; Significant = p<. 01; NS = Not significant.

To establish firm evidence for X-linked transmission of affective disorder would require full genealogical analysis of the affected families with knowledge of presence or absence of affective disorders in both male and female subjects over three or four generations—evidence that we do not have. Our findings do indicate, however, that for unknown—dare I say mysterious—reasons, early death of maternal grandfathers predicts an increased incidence of affective disorder in the grandsons. Still more exciting is that long-lived maternal grandfathers predict unusual psychological stability in their grandsons—evidence that positive mental health may be in part genetic. The association of very long-lived maternal grandfathers with men scoring low on the NEO anxiety score is particularly intriguing.

My own guess is that someday soon an accomplished geneticist with a larger study, better characterized maternal grandfathers, and complete DNA analyses will win the Nobel Prize for explicating this phenomenon. At present it must be considered a preliminary finding, of interest only to the curious. Still, it is provocative, and would not have been discovered but for 60 years of follow-up. And it's a perfect example of how a seductive gleam can wink out unexpectedly from unruly heaps of unsorted longitudinal data, to be revealed eventually by Time's assay as either brass or true gold.